



CLIMATEWINS WEATHER PATTERNS AND CLIMATE CHANGE ANALYSIS

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PROJECT OVERVIEW

01

ClimateWins, a European nonprofit organization, is implementing machine learning to better understand historical weather patterns and predict the future impacts of climate change across the globe.

02

The purpose of this project is to evaluate machine learning methods and determine their effectiveness in identifying weather trends, categorizing climate conditions and predicting future weather outcomes.

03

This analysis supports ClimateWins' long-term strategy to use data-driven insights to improve climate preparedness, inform planning efforts and strengthen its ability to respond to changing environmental conditions.

PROJECT OBJECTIVE

To assess and implement machine learning techniques that analyze historical European weather data to identify climate patterns and develop predictive models that support accurate forecasting and strategic climate planning.



HYPOTHESIS



Historical Weather
Data Contains
Detectable Climate
Trends



Extreme Weather
Events Have
Increased in
Frequency



Prediction accuracy
will vary between
machine learning
algorithms due to
differences in their
ability to model
complex climate
patterns

DATASET OVERVIEW

DataSet Characteristics

- Data from 18 European Weather Stations
- Time Range: Late 1800s to 2022
- Daily Weather observations
- Collected by the European Climate Assessment & Dataset Project (ECA & D)

Key Climate Variables Used

- Temperature
- Wind Speed
- Snowfall
- Global radiation





DATA BIAS AND LIMITATIONS

Historical Data Limitations

Older records may contain measurements inconsistencies due to less advanced technology, manual recording methods and inconsistent measurements.

Regional Bias

Weather stations with more complete data may influence model training more heavily, resulting in uneven representation across geographic areas.

Temporal Bias

Changes over time in measurement methods, technology or environmental conditions may introduce inconsistencies.

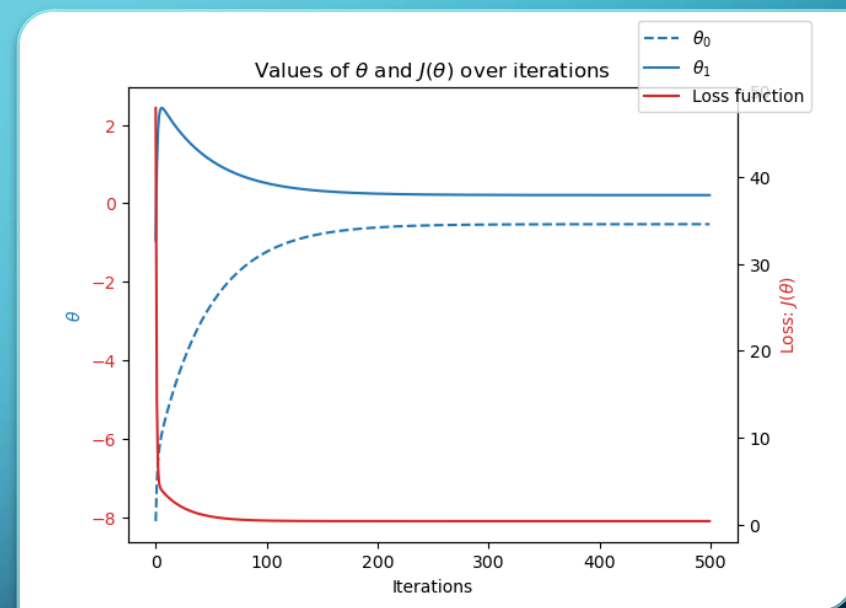
DATA OPTIMIZATION

Gradient Descent was used to optimize the machine learning model by minimizing prediction error and identifying the most accurate parameter value.

- The red line represents the model's predictions error.
- The Blue line represents the model's parameter adjusting over time.

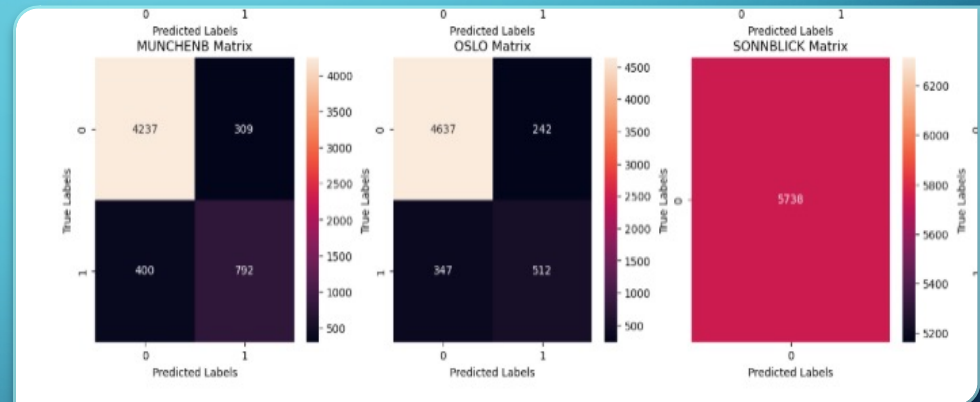
The decreasing loss and stabilizing parameter values confirm that the model was successfully optimize. This ensures:

- Improved prediction accuracy
- Reliable model performance
- Strong foundation for climate prediction analysis



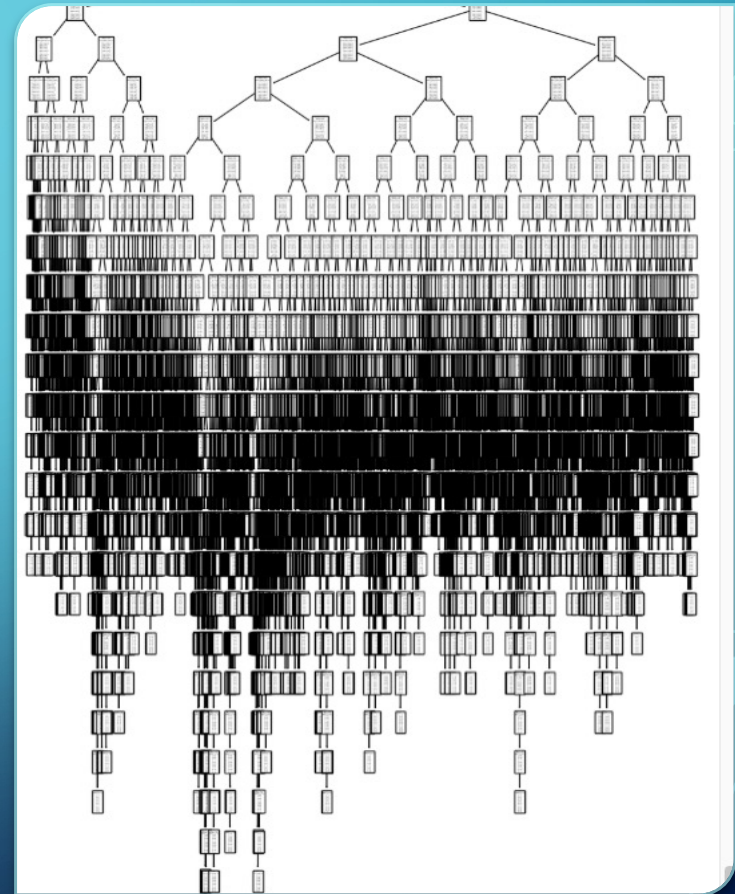
ALGORITHM 1: K-NEAREST NEIGHBOR (KNN)

- KNN was used to classify new data points based on their similarity to already stored data. It operates on the principle that “birds of a feather flock together”.
- Key Characteristics:
 - Classifies data based on similarity
 - Simple and effective model
 - Performs well with structured datasets
 - High prediction accuracy
 - Easy to Implement
- Based on the image to the right, Sonnblick, stands out. The high accuracy is misleading and suggests underfitting due to imbalance.
- The estimated accuracy ranges from 85% to 95%.



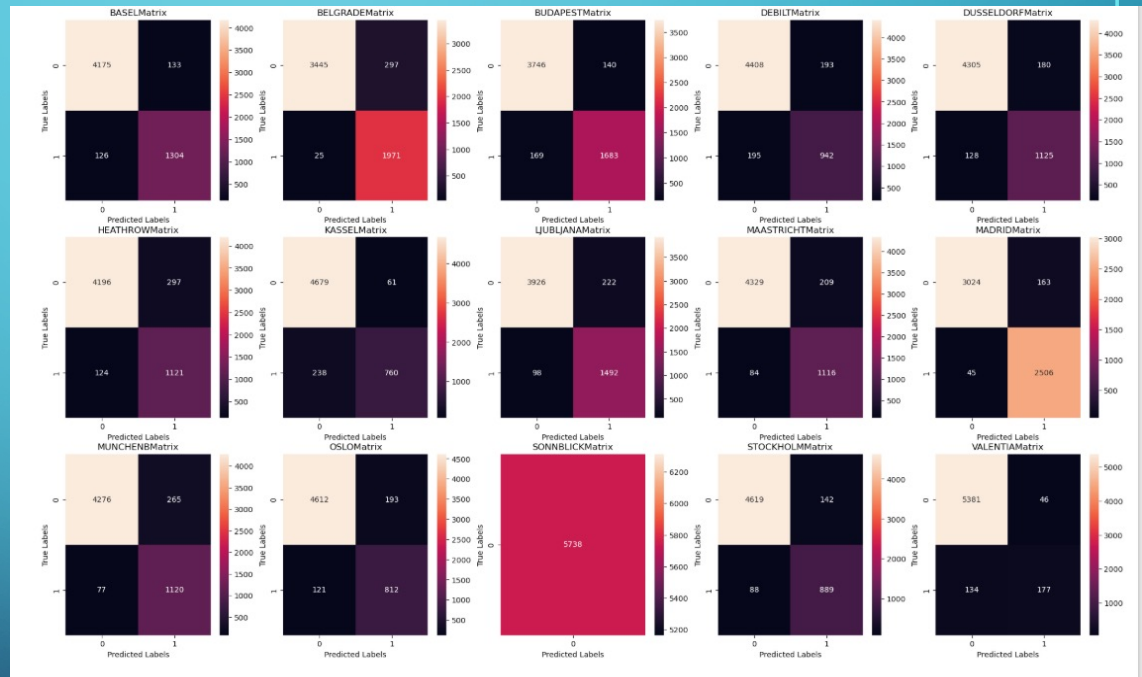
ALGORITHM 2: DECISION TREE

- A decision tree makes predictions by following a sequence of questions based on the features of the dataset presented.
- It splits data into smaller and smaller groups based on the decision rules until it finally reaches a prediction.
- Key Characteristics:
 - Capture Complex Relationships
 - High Predictive Power
 - Handles Mixed Data Types
 - Increased Computational Cost
 - Potential Instability
- The average prediction for the ClimateWins data is 60%.
- The model presented demonstrates strong pattern recognition, however, it reveals high complexity and deep branching which increases overfitting risk and reduces interpretability.



ALGORITHM 3: ARTIFICIAL NEURAL NETWORK

- An Artificial Neural Network (ANN), is a machine learning model inspired by the structure of the human brain.
- Each Layer contains nodes called neurons. These neurons apply weights and activation functions to learn patterns in the data.
- Key Characteristics:
 - Captures complex nonlinear relationships.
 - Handles high dimensional data well.
 - Learns intricate patterns automatically
 - Computationally it is more expensive.
 - Can overfit if not regularized.
- Based on the chart provided, the best achieved accuracy is 62%.



RECOMMENDATION

- Based on the Algorithms and models used, I would recommend the ANN model for ClimateWins to use. The confusion matrices show high correct classification rates across regions. It also handles nonlinear weather relationships.

NEXT STEPS



Broaden Geographic Representation

Incorporating additional weather stations and expanding regional coverage would improve the model's ability to generalize across diverse climate conditions. A wider footprint reduces regional bias and increases confidence in predictive reliability.



Implement Ongoing Model Governance

Climate data evolves over time.

Establishing a retraining framework and performance monitoring process will ensure models remain relevant as weather patterns shift.



Strengthen Data Integrity

Historical climate records often contain inconsistencies due to measurement changes, equipment variation or recording gaps. A structured data auditing process and standardization preprocessing protocols would improve signal quality and reduce modeling noise.

QUESTIONS?

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 - Github: <https://github.com/tbarnett123/ClimateWins-MachineLearning-Portfolio>